

Pham Ngoc Gia Bao

also known as Alëksis Arendt

Undergraduate Student in Mathematics

University of Science, Vietnam National University Ho Chi Minh City (HCMUS)

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Education

B.Sc. in Mathematics

2025–present

University of Science, Vietnam National University Ho Chi Minh City (HCMUS)

Nguyen Dinh Chieu Gifted High School, Dong Thap

2022–2025

Mathematics-specialized track.

Relevant coursework: Linear Algebra, Calculus, Discrete Mathematics, Programming, Physics.

Independent study: Real Analysis, Measure Theory, Differential Geometry.

Academic Interests

Classical Analysis, Geometric Measure Theory, Harmonic Analysis, Incidence Geometry, and Discrete Geometry.

Awards and Distinctions

- **Bronze Medal**, April 30 Traditional Olympic Competition in Mathematics, Vung Tau, Vietnam.
- **Second Prize**, VIASM Mathematics Training Camp, Da Lat, 2024.
- **Ranked 1st**, Dong Thap selection for the National Mathematics Competition.

Selected Mathematical Writing and Projects

Personal mathematical blog and expository writings

Blog

Maintained a personal academic blog for organizing mathematical notes, selected writings, and small academic projects, mainly in analysis and geometry.

Expository report on Newtonian ODE models

Manuscript in preparation

Studied a one-dimensional Newtonian model involving gravity, drag, and feedback control; analyzed damping regimes, stability, and energy estimates; implemented numerical simulations using Python.

Forthcoming expository notes

In preparation

Topics include the polynomial method, billiard problems, and Lagrangian–Hamiltonian mechanics.

Selected Software Projects

arXiv Reader

Source code

Built a local arXiv research tool using Python and Scrapy, with category-based queries, keyword ranking, and JSON snapshots.

NDC Multimedia Club website

Source code | Demo

Built and deployed a website for the Multimedia Club at Nguyen Dinh Chieu Gifted High School using React.js, MUI, and Netlify.

Technical Skills

LaTeX, Python, C++, Git, Markdown/MDX.

Languages

Vietnamese: native.

English: academic working proficiency.

German: currently learning.